

Lecture

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When Control Design Serves Human Assistance and Rehabilitation with Exoskeletons

Abstract

Assisting people with disabilities and older adults in their daily activities through wearable robotic technologies has attracted growing interest over the past decades. This interest is driven by the increasing prevalence of mobility impairments and the rapid aging of the global population. Rising life expectancy, combined with declining birth rates, is expected to further accelerate population aging and increase the demand for effective, safe, and user-centered assistive technologies. At the same time, rapid advances in portable technologies, including compact and miniaturized components, low-power electronics, lightweight actuators, and increasingly accessible sensing technologies, are improving the wearability, autonomy, and affordability of assistive devices. Human-wearable technologies such as exoskeletons, wearable biosensors, and functional electrical stimulation systems are progressively moving from research prototypes toward practical applications. This evolution has fostered increasingly interdisciplinary research at the intersection of mechanical design, sensing, actuation, robotics, human–robot interaction, and control. In this context, control plays a central role in enabling wearable robots to interact safely, intuitively, and effectively with the human body. This talk will review major challenges and recent advances in wearable robotics, with a particular focus on the control of assistive and rehabilitation exoskeletons. Different control strategies developed to provide physical assistance and support motor rehabilitation will be presented. Their performance will be illustrated through numerical simulations and real-time experimental demonstrations involving human subjects.